

This document provides additional assistance with wiring your Extron IPL PRO enabled product to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instruction, refer to the user's manual for the specific Extron IPL PRO enabled product or the controlled device manufacturer supplied documentation.

Device Specifications:

Device Type: Video Projector
Manufacturer: BenQ
Firmware Version: N/A
Model(s): MX602, MW603

Tested on the Following Software and Firmware Versions:

IP Link Pro Control Processor Firmware	Touch Link Panel Firmware	GS Version
2.04.0000-b007	2.01.0001-b001	1.2.2

Version History

Module Version	Date	Notes
1_0_0_0	1/30/2017	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'. Example:
InterfaceName.Unidirectional = 'True'
- connectionCounter variable must be set the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15. Example:
InterfaceName.connectionCounter = 5

Control Commands, States, and Qualifiers:

3DSync	'Sync Off'	'Top Bottom'	'Frame Sequential'
AspectRatio	'Auto' '16:9'	'Real' '16:10'	'4:3'
AudioMute	'On'	'Off'	
AutoImage	None		
Freeze	'On'	'Off'	
Input	'Computer/YPbPr' 'Composite' 'USB Display'	'Computer 2/YPbPr 2' 'S-Video' 'USB Reader'	'HDMI' 'Network'
LampMode	'Normal' 'Smart Eco 2'	'Eco' 'Smart Eco 3'	'Smart Eco'
MenuNavigation	'Up' 'Right' 'Menu Off'	'Down' 'Enter'	'Left' 'Menu On'
PictureMode	'Dynamic' 'Cinema'	'Presentation' 'User 1'	'sRGB' 'User 2'
Power	'On'	'Off'	
VideoMute	'On'	'Off'	
Volume	'Up'	'Down'	

Status Available:

3DSync	'Sync Off'	'Top Bottom'	'Frame Sequential'
AspectRatio	'Auto' '16:9'	'Real' '16:10'	'4:3'
AudioMute	'On'	'Off'	
ConnectionStatus	'Connected'	'Disconnected'	
Freeze	'On'	'Off'	
Input	'Computer/YPbPr' 'Composite' 'USB Display'	'Computer 2/YPbPr 2' 'S-Video' 'USB Reader'	'HDMI' 'Network'
LampMode	'Normal' 'Smart Eco 2'	'Eco' 'Smart Eco 3'	'Smart Eco'
LampUsage	Hours		
Lamp2Usage	Hours		
PictureMode	'Dynamic' 'Cinema'	'Presentation' 'User 1'	'sRGB' 'User 2'
Power	'On'	'Off'	
VideoMute	'On'	'Off'	
VolumeStatus	0 – 10		

Cable and Adapter Requirements:

Captive Screw to Female DB9 RS-232 Serial Cable

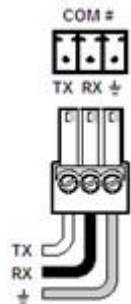
Notes for the Device:

Serial communication:

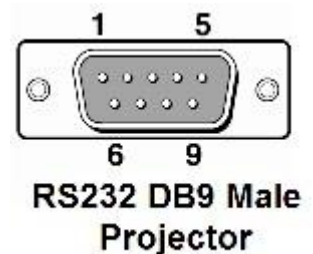
Port Type: RS-232
Baud Rate: 115200
Data Bits: 8

Parity: None
Stop Bits: One
Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD	→	2	RxD
RxD	←	3	TxD
GND		5	GND



General Notes:

Network communication:

When configuring the Ethernet interface, be sure the device settings match that of the GS interface settings.

Port Type:	Ethernet
Default Port:	8000
Logon Credentials Supported:	No
Multi-Connection Capabilities:	Undetermined
Port Changeability:	Yes

Ethernet Module Configuration Description:

Please refer to user manual for settings and changes to the network communication

Notes for the Device:

Appendix A. Set Commands

3D Sync Frame Sequential	\x0D*3d=fs#\x0D
3D Sync Sync Off	\x0D*3d=off#\x0D
3D Sync Top Bottom	\x0D*3d=tb#\x0D
Aspect Ratio 16:10	\x0D*asp=16:10#\x0D
Aspect Ratio 16:9	\x0D*asp=16:9#\x0D
Aspect Ratio 4:3	\x0D*asp=4:3#\x0D
Aspect Ratio Auto	\x0D*asp=AUTO#\x0D
Aspect Ratio Real	\x0D*asp=REAL#\x0D
Audio Mute Off	\x0D*mute=off#\x0D
Audio Mute On	\x0D*mute=on#\x0D
Auto Image None	\x0D*auto#\x0D
Freeze Off	\x0D*freeze=off#\x0D
Freeze On	\x0D*freeze=on#\x0D
Input Composite	\x0D*sour=vid#\x0D
Input Computer 2/YPbPr 2	\x0D*sour=RGB2#\x0D
Input Computer/YPbPr	\x0D*sour=RGB#\x0D
Input HDMI	\x0D*sour=hdmi#\x0D
Input Network	\x0D*sour=network#\x0D
Input S-Video	\x0D*sour=svid#\x0D
Input USB Display	\x0D*sour=usbdisplay#\x0D
Input USB Reader	\x0D*sour=usbreader#\x0D
Lamp Mode Eco	\x0D*lampm=eco#\x0D
Lamp Mode Normal	\x0D*lampm=lnor#\x0D
Lamp Mode Smart Eco	\x0D*lampm=seco#\x0D
Lamp Mode Smart Eco 2	\x0D*lampm=seco2#\x0D
Lamp Mode Smart Eco 3	\x0D*lampm=seco3#\x0D
Menu Navigation Down	\x0D*down#\x0D
Menu Navigation Enter	\x0D*enter#\x0D
Menu Navigation Left	\x0D*left#\x0D
Menu Navigation Menu Off	\x0D*menu=off#\x0D
Menu Navigation Menu On	\x0D*menu=on#\x0D
Menu Navigation Right	\x0D*right#\x0D
Menu Navigation Up	\x0D*up#\x0D
Picture Mode Cinema	\x0D*appmod=cine#\x0D
Picture Mode Dynamic	\x0D*appmod=dynamic#\x0D
Picture Mode Presentation	\x0D*appmod=preset#\x0D
Picture Mode sRGB	\x0D*appmod=srgb#\x0D
Picture Mode User 1	\x0D*appmod=user1#\x0D
Picture Mode User 2	\x0D*appmod=user2#\x0D
Power Off	\x0D*pow=off#\x0D
Power On	\x0D*pow=on#\x0D
Video Mute Off	\x0D*blank=off#\x0D
Video Mute On	\x0D*blank=on#\x0D
Volume Down	\x0D*vol=-#\x0D
Volume Up	\x0D*vol=+#\x0D

Appendix B. Query Commands

3D Sync	\x0D*3d=?#\x0D
Aspect Ratio	\x0D*asp=?#\x0D
Audio Mute	\x0D*mute=?#\x0D
Freeze	\x0D*freeze=?#\x0D
Input	\x0D*sour=?#\x0D
Lamp Mode	\x0D*lampm=?#\x0D
Lamp Usage	\x0D*ltim=?#\x0D
Lamp2 Usage	\x0D*ltim2=?#\x0D
Picture Mode	\x0D*appmod=?#\x0D
Power	\x0D*pow=?#\x0D
Video Mute	\x0D*blank=?#\x0D
Volume Status	\x0D*vol=?#\x0D